Novel region-based modeling for human detection within highly dynamic aquatic environment

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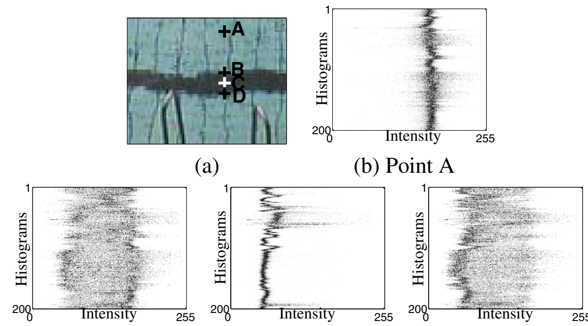
Abstract

One difficult challenge in autonomous video surveillance is on handling highly dynamic backgrounds. This difficulty is compounded if foreground objects of interest are partially hidden by specular reflections or glare. In this paper, we provide numerous insights into the technical difficulties faced in developing an automated video surveillance system within a hostile environment: an outdoor public swimming pool. For robust detection performance, we focused on two central aspects: i) effective modeling of the dynamic outdoor aquatic background with rapid illumination changes, splashes and random spatial movements of background elements, owing to the movement of water ripples; and ii) enhancing the visibility of swimmers that are partially hidden by specular reflections. Several innovations have been introduced from scratch in this paper. The first is the development of a scheme that models the background as regions of dynamic homogeneous processes. This model facilitates the implementation of an efficient spatial searching scheme for background subtraction that could exploit long-range spatial dependencies between pixels. The second is the implementation of a spatio-temporal filtering scheme that enhances the detection of swimmers that are partially hidden by specular reflections of artificial nighttime lighting, serving as a pre-processing module to foreground detection for nighttime operation. These various algorithms have been tightly integrated under a unified framework and demonstrated on a busy Olympic-sized outdoor public pool.

1 Introduction

Recent research efforts in automated surveillance have progressively been driven by the motivation to operate in real environments [1]-[5] for applications such as homeland security, moving away from controlled laboratory settings. There is hence a crucial need for efficient and robust algorithms that are highly stable within dynamic real environments and produce sufficient quality of intermediate representations for high-level tasks.

This paper provides various insights into the technical difficulties faced in developing an automated video surveillance system within a hostile environment: an outdoor public swimming pool. Owing to the random movement of water ripples and splashes common at aquatic environments, the system needs to operate within a dynamic background with significant background pixel movements as depicted in Figure 1. The horizontal axis of the figure represents the gray level intensities of pixels while each row in the vertical axis represents the intensity histogram from a pixels over 200 frames. Subsequent rows depict the histogram of that pixel over the next 200 frames, hence capturing the pixel's intensity patterns over a total of 40,000 frames. The wide distribution of the intensity histogram demonstrates a higher level of complexity and challenges involved in such considered problem compared with the problem of detecting human on the ground that has a much less distributed intensity histogram as shown in Figure 6 of [3].



(a) (b) Point A
(c) Point B (d) Point C (e) Point D

Figure 1: Portion of a typical pool: Intensity histogram of pixels over time. Wide distributed histograms due to rapid movements of pixels belonging to elements of the background.

For robust detection, we focused on two central aspects: i) effective modeling the dynamic outdoor aquatic background that experiences rapid illumination changes, splashes and random spatial movements of background elements, and ii) enhancing the visibility of swimmers that are partially hidden by specular reflections and glare.

We observed that dynamic backgrounds typically carry

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the following properties: i) illumination variation over time due to (possibly rapid) changes in ambient lighting, and ii) random and rapid spatial movements of background elements. We found that an effective way of modeling such dynamic backgrounds is to represent it as a composition of dynamic background patches. The background is first divided into a series of non-overlapping blocks. Clustering of pixel colors within each block is then performed which produces a small number of homogeneous regions, each of which is modeled using a multivariate Gaussian distribution. As will be explained later, this model facilitates the implementation of an efficient spatial searching scheme for background subtraction that exploits the long-range spatial dependencies between pixels.

Developing a real system with long operational hours meant that we needed to overcome problems caused by specular reflections of artificial nighttime lighting due to the non-optimal position of the video cameras. These glare-like effects have severely reduced the visibility of swimmers in the pool. A spatio-temporal filtering scheme is thus introduced for frame enhancement before foreground detection to recover pixels of swimmers that are partially hidden by these specular reflections. Based on a newly defined concept termed motion frequency, pixels are categorized and then adaptively channeled into appropriate spatio-temporal and color compensation filters for frame enhancement.

This paper is organized as follows: Section 2 compares related work and puts our work into perspective. Section 3 describes the methodology in detail, illustrating the following innovations: i) a tight-coupled region-based background modeling and thresholding operation that combines thresholding-with-hysteresis, denoising and connected component grouping to accurately detect and track swimmers amid noisy and highly dynamic aquatic environments, and ii) a night-time processing strategy comprising a unique filter switching scheme to effectively recover swimmers hidden under the glare of specular reflections. Experimental results and concluding remarks are presented in Sections 4 and 5 respectively.

2 Previous work

Background subtraction has been a popular framework for recovering foreground regions from video sequences captured using stationary cameras. However, one major challenge for background subtraction lies in the difficulty to construct effective background models. Additionally, this difficulty is further compounded when considering highly dynamic environments. In the following, we review several related work that are relevant to our algorithm to put our efforts in perspective.

Pixel-based modeling has been a popular approach among the literature. For instance, the Pfinder system [6] demonstrated real time tracking of single person in an indoor environment by constructing each background pixel based on a single Gaussian distribution in YUV space. Such method is typically successful when applied to controlled indoor environments with relatively stationary backgrounds. Similar work using Kalman filter for background updating by [7] shares the same limitation, showing fundamental limitation of these methods for applying on problem with highly dynamic background.

Recent efforts by [2],[8] proposed the use of a mixture of Gaussian (MoG) distribution being a generalization to the mentioned single Gaussian method for outdoor surveillance. The rely on an accurate classification process to correctly update the Gaussian mixtures is a practical concern for this method. Approximation of the MoG distributions using an online version of the k-means algorithm would naturally reduces the accuracy of background modeling.

Besides the pixel-based approach, the classic paper by [9] introduced a block-based method for background modeling, where each image block was fitted to a second-order bivariate polynomial. However, the statistical framework of this method that necessitates the use of an image block has constrained the detection to be in block-based unit. As a result, this method has been more for coarse detection.

Our method differs from existing approaches in which the background is modeled as a set of block-derived region-based units as opposed to pure pixel-based or block-based ones. Each region is assumed non-stationary for describing disturbance at the background, and experiences changes in illumination and color due to different ambient conditions. Additionally, the approach also complements an efficient spatial searching scheme for background subtraction that exploits the long-range spatial dependencies between pixels within a dynamic background.

3 The proposed methodology

Figure 2 presents the architecture of our methodology for robust foreground detection within highly dynamic environments. It consists of a novel region-based background modeling method that builds background model of a considered scene and a robust detection strategy which is based on thresholding-with-hysterisis principle within a background subtraction framework. Finally we describe a spatio-temporal filtering scheme that enhances foreground detection amid specular reflections of artificial night-time lighting.

3.1 Region-based background modeling

In learning the region-based background model, consider that a sequence of $N_1 \times N_2$ background frames are available, and each frame background frame is divided into $n_1 \times n_2$ non-overlapping square blocks with size of $s \times s$ for each, where $n_1 = (N_1/s)$ and $n_2 = (N_2/s)$. Let $\boldsymbol{X}_{a,b} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p\}$ be pixels collected from square

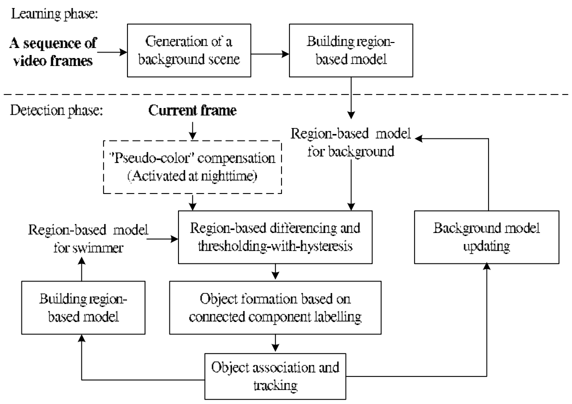

Figure 2: Architecture of the proposed algorithm.

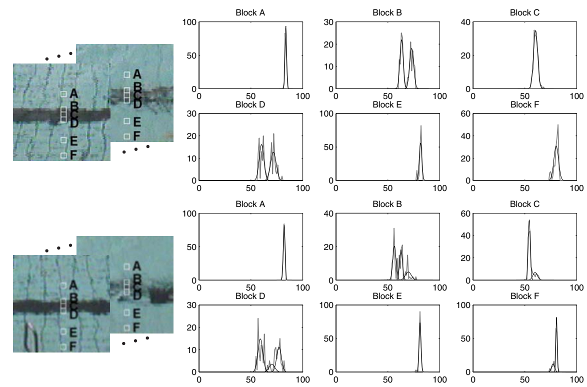

Figure 3: Luminance histograms of pixel vectors $\boldsymbol{X}_i$, $i \in \{\mathrm{A}, \ldots, \mathrm{F}\}$ extracted from Blocks A to F as marked at two typical example images on the left, showing multiple peaks when more than one objects projected onto the block over time. Modeling each homogeneous regions as one single-Gaussian distribution demonstrates good approximation of the original signal.

blocks of these background frames with position index-(a, b), where $1 \leq a \leq n_1$ and $1 \leq b \leq n_2$. To establish a set of homogeneous region models characterizing the dynamic property of background, we perform background segmentation by clustering every $\boldsymbol{X}_{a,b}, \forall\, a, b$ that groups each $\boldsymbol{X}_{a,b}$ into homogeneous regions $\{\boldsymbol{R}^1_{a,b}, \ldots, \boldsymbol{R}^c_{a,b}\}$.

Figure 3 illustrates a study of luminance histogram of pixels collected from different square blocks as marked at images on the left. Multiple peaks are observed when there are more than one background element within the same block over time, such as for the cases of blocks B and D which contain both elements of water and the lane divider. It is therefore reasonable to model each peak $\boldsymbol{R}^k_{a,b}$ as a single multivariate Gaussian (in the 3-D color space). The composition of which forms a mixture of Gaussian-like distributions for each block. In other words, clustering produces the effect of breaking down the complicated pixel color processes of a block into homogeneous region processes with each could be reasonably modeled using a single multivariate Gaussian distribution.

Let's denote the mean and covariance matrix of a homogeneous region $\boldsymbol{R}^k_{a,b}$ by $\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}} = \{\mu^1_{\boldsymbol{R}^k_{a,b}}, \ldots, \mu^d_{\boldsymbol{R}^k_{a,b}}\}$ and Σ, respectively, where $d = 3$ is the dimension of the color space. Thus, using probabilistic notation, the probability that a pixel $\boldsymbol{x}_{i,j} = \{x^1_{i,j}, \ldots, x^d_{i,j}\}$ belongs to a homogeneous background region $\boldsymbol{R}^k_{a,b}$ is given by:

$$P(\boldsymbol{x}_{i,j} | \boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \times \exp\Big\{ -\frac{1}{2}(\boldsymbol{x}_{i,j} - \boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}})^T \Sigma^{-1} (\boldsymbol{x}_{i,j} - \boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}) \Big\}. \quad (1)$$

Without losing much accuracy at the final segmentation, it is commonly to assume pixel values in different dimensions are independent to reduce computational complexities. Based on such assumption, we further simplify the probability measure in (1) to be:

$$P(\boldsymbol{x}_{i,j} | \boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}) = \prod_{l=1}^{d} \frac{1}{\sqrt{2\pi}\sigma^l_{\boldsymbol{R}^k_{a,b}}} \exp\Big\{ -\frac{(x^l_{i,j} - \mu^l_{\boldsymbol{R}^k_{a,b}})^2}{2(\sigma^l_{\boldsymbol{R}^k_{a,b}})^2} \Big\}, \quad (2)$$

where $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}} = \{\sigma^1_{\boldsymbol{R}^k_{a,b}}, \ldots, \sigma^d_{\boldsymbol{R}^k_{a,b}}\}$ is the standard deviation of $\boldsymbol{R}^k_{a,b}$. Such assumption is reasonable as we have chosen the CIELa*b* color space that the luminance component (i.e., L component) of pixel signal is well decorrelated from its chronimance component (i,e, a* and b* components). Finally, the similarity of a pixel with respect to a homogeneous region of the background could thus be quantified mathematically using the distance measure below which is related to the negative log-likelihood of (2):

$$D(\boldsymbol{x}_{i,j} | \boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}) = \sqrt{\sum_{l=1}^{d} (x^l_{i,j} - \mu^l_{\boldsymbol{R}^k_{a,b}})^2 / (\sigma^l_{\boldsymbol{R}^k_{a,b}})^2}. \quad (3)$$

Most often (3) will yield a small value for background pixels and vice versa for foreground swimmers. Thus, (3) provides an effective and compelling metric to devise a robust thresholding scheme for foreground detection. The normalization by the denominator in (3) provides an adaptation to changing dynamicity of background regions over time as facilitated through updated values of $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}$.

Such proposed region-based background modeling scheme has enabled dynamic backgrounds to be represented as compositions of actively moving, homogeneous background regions of arbitrary size and shape. As will be described in Section 3.2, this approach would facilitate an efficient comparison of a pixel's color with background models over significant distances covering neighboring blocks.

In the following, we describe the two essential steps when building a live system that comprises: i) building an initial background model with the possibly that there is foreground in the frames captured during the learning phase and ii) recursively updating of the established background model during the online operation.

3.1.1 Learning initial background model

Video frames captured may contain both moving and stationary foreground swimmers. Thus, the first essential step is to build a sequence of "clean" background frames for the background formation process. We first apply a skin color model to isolate swimmer pixels from the background formation process. Residue swimmer pixels are then removed using a temporal vector median filter.

Let $\boldsymbol{X}_{i,j}$ be an array of color vectors collected over T number of frames, i.e. $\boldsymbol{X}_{i,j} = \{\boldsymbol{x}^t_{i,j} \mid t = 1, \ldots, T\}$, where $\boldsymbol{x}^t_{i,j}$ is the color vector of the tth image at position-(i, j). The sampling rate that determines the temporal interval between two $\boldsymbol{x}^t_{i,j}$ was decided empirically after considering a tradeoff between the duration needed for the learning phase and the efficiency to remove swimmer pixels. A clean background frame $\boldsymbol{B}_1$ is obtained by performing vector median filtering on $\boldsymbol{X}^t_{i,j}$, $\forall\ i, j$, which generates:

$$\boldsymbol{B}_1 = \Big\{\boldsymbol{y}_{i,j} \mid i = 1, \ldots, N_1 \text{ and } j = 1, \ldots, N_2\Big\}\,, \qquad (4)$$

where $\boldsymbol{y}_{i,j}$ is obtained based on

$$\boldsymbol{y}_{i,j} = \Big\{\boldsymbol{x}^q_{i,j} \in \boldsymbol{X}_{i,j} \mid \min_{\boldsymbol{x}^q_{i,j}} \sum_{p=1}^{T} |\boldsymbol{x}^p_{i,j} - \boldsymbol{x}^q_{i,j}|\Big\}\,. \qquad (5)$$

The subsequent background frames $\{\boldsymbol{B}_l | l = 2, \ldots, N\}$ are also generated based on the same operation by maintaining an overlapping sliding window that captures pixel vector $\boldsymbol{X}_{i,j} = \{\boldsymbol{x}^t_{i,j} \mid t = l, \ldots, T + l\}$.

From our experiments, we found CIELa*b* to produce better swimmer segmentation results compared to other color spaces. To establish the proposed background model, $\{\boldsymbol{B}_l \mid l = 1, \ldots, N\}$ are first converted into the CIELa*b* space, forming matrix $\{\boldsymbol{B}'_l \mid l = 1, \ldots, N\}$. Subsequently, the following two steps are performed to construct a set of region-based background models:

i) Dividing each $\boldsymbol{B}'_l$ into $n_1 \times n_2$ of non-overlapping $s \times s$ square blocks, and
ii) Applying a hierarchical k-means on each square block to obtain cluster centers $\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}$ and standard deviations $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}$ of homogeneous regions $\boldsymbol{R}^k_{a,b}$. The clustering process on each $\boldsymbol{X}_{a,b}$ starts by initiating each pixel vector to be one dominant data cluster. In the subsequent iterations, smaller and more compact data clusters are formed through splitting until the distance of any two closest cluster centers is smaller than a threshold or the number of clusters reaches c, a pre-determined value, whichever is achieved first. Therefore, this yield the initial background model to be

$$\boldsymbol{C}_{a,b} = \Big[\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}\,,\, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}\Big]^T, \text{ where } k = 1, \ldots, c. \qquad (6)$$

Since we have assumed each color channel is independent, parameters $\boldsymbol{\mu}_{\boldsymbol{R}^k_{a,b}}$ and $\boldsymbol{\sigma}_{\boldsymbol{R}^k_{a,b}}$ are obtained by computing the means and standard deviations in the respective color channel.

3.1.2 Updating background model parameters

There is a need to constantly and recursively update the background model in order to accurately reflect the environmental conditions, in view of rapid illumination changes due to the sudden blocking of the sun by clouds, violent ripple movements on the water surface, etc. Thus, we update parameters $\boldsymbol{\sigma}^t_{\boldsymbol{R}^k_{a,b}}$ and $\boldsymbol{\mu}^t_{\boldsymbol{R}^k_{a,b}}$ based on the following scheme:

$$\boldsymbol{\mu}^t_{\boldsymbol{R}^k_{a,b}} \leftarrow (1-\rho)\boldsymbol{\mu}^{t-1}_{\boldsymbol{R}^k_{a,b}} + \rho\boldsymbol{\mu}^t_{\boldsymbol{R}^k_{a,b}}\,, \qquad (7)$$

$$\boldsymbol{\sigma}^t_{\boldsymbol{R}^k_{a,b}} \leftarrow (1-\rho)\boldsymbol{\sigma}^{t-1}_{\boldsymbol{R}^k_{a,b}} + \rho\boldsymbol{\sigma}^t_{\boldsymbol{R}^k_{a,b}}\,, \qquad (8)$$

where $\rho = 1/T$ is the learning factor for adapting current changes, with T being defined earlier as the number of frames collected to construct the initial clean background sequence. This method provides a recursive linear scheme for continuously updating background models, and shares some similarity with the approach used in [2].

In addition, the ability to create new models and destroy unnecessary models is crucially needed. Each homogeneous region background model will be eliminated if no pixels are classified under its label after T frames. Creating new models is meanwhile essential to represent newly emerged background regions which have not been captured during the initial learning process. New Gaussian models will thus be created for background pixels that are ill-matched to any background models and collectively satisfy a homogeneity criterion to be statistically categorized as new regions.

3.2 Foreground swimmer detection and modeling

The main problem for foreground detection within highly dynamic environments is to attain a good tradeoff between maximizing target detection while minimizing foreground false positives. In this section, we describe a scheme that balances both requirements based on a modified background subtraction framework that incorporates active background movements, combining the principles of thresholding-with-hysteresis, denoising and connected component grouping.

Let $\boldsymbol{x}_{i,j}$ be a color pixel at location-(i, j) within the square block-(a, b). Using (3) and considering the pixel's possible movements within a local window, we define the color discrepancy between the pixel and the background as:

$$D^{\min}_{i,j} = \min\big\{D(\boldsymbol{x}_{i,j}|\boldsymbol{\mu}_{\boldsymbol{R}^k_{a+r,b+r}}, \boldsymbol{\sigma}_{\boldsymbol{R}^k_{a+r,b+r}})\big\}\,, \qquad (9)$$

where the dimension of the local window is $(2r + 1) \times (2r + 1)$ in square block unit. In other words, we are computing the minimum distance metric between the pixel and

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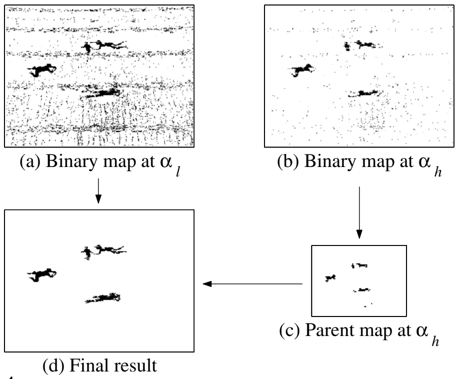

Figure 4: Thresholding for foreground swimmer detection that combines the principle of thresholding-with-hysteresis, denoising and connected component grouping.

all background blocks within the local window. Computing $D_{i,j}^{\min}$ $\forall\, i, j$ yields a difference image for a considered frame against the background. Intuitively, a swimmer could then be recognized by the fact that its pixels produce statistically large $D_{i,j}^{\min}$ values. A binary segmentation map, $M_{i,j}$ is then generated through thresholding:

$$M_{i,j} = \begin{cases} 0 \text{ background} & , \quad D_{i,j}^{\min} < \alpha, \\ 1 \text{ foreground} & , \quad \text{otherwise}. \end{cases} \qquad (10)$$

This generates a binary map with "0"s and "1"s marking background and foreground, respectively.

Such consideration of neighboring blocks in computing $D_{i,j}^{\min}$ has significantly reduced mis-classification errors at problematic regions with significant background movements such as seemingly moving shadows caused by water refractions and areas around lane dividers. The increase in search space effectively incorporates background movements across various situations, hence reducing false positives at the final results. The choice of an optimum threshold has always been a tricky problem within background subtraction frameworks. In the following, a better solution that combines the principles of thresholding-with-hysteresis, denoising and connected component grouping is developed as a superior alternative to the intuitive single threshold scheme.

3.2.1 Choice of threshold, α

We suggest that thresholding be performed in a hierarchical manner, which has been proven of higher efficiency as reported in earlier work [3], based on a strategy as shown in Figure 4. Apart from producing a full resolution binary map on $D_{i,j}^{\min}$ using a low threshold α_l, we also construct a coarser resolution of the binary map, termed the *parent map*, obtained from using a high threshold α_h. Elements in the coarse resolution parent map are labeled "1" only if there is a majority of corresponding binary pixels at the full resolution with $D_{i,j}^{\min} > \alpha_h$.

Thus, a foreground is formed if a connected region of pixels has $D_{i,j}^{\min}$ above α_l with the added requirement that its corresponding area in the parent map must also form a connected blob. Comparing the images in (b) and (c), the use of this thresholding-with-hysterisis approach clearly achieves the best of both-worlds, enhanced detection as well as a lower number of false positives.

3.2.2 Updating swimmer models

In the same way as forming the background model, once a swimmer is detected, a region-based swimmer model is also defined and updated every frame to improve the segmentation accuracy. During online processing phase with the establishment of foreground model, the color discrepancy measure between pixels' color of the incoming image and swimmer cluster centroids $\mathbf{C}^f(i,j)$ within its corresponding and also the respective surrounding eight connected swimmer blocks are computed, giving

$$\mathbf{D}^f = \left\{ D_{i,j}^{f\min} \mid i = 1, \ldots, M \; ; \; j = 1, \ldots, N \right\}. \qquad (11)$$

Thus, pixels with $D_{i,j}^{f\min} < D_{i,j}^{\min}$ are firstly classified as the highest confidence swimmer pixels. The remaining swimmers pixels are detected using the thresholding-with-hysteresis scheme detailed earlier. Any thresholding algorithm generally exhibits the limitation to detect pixel of foreground that has a close similarity to the color of background, while trying to maintain a low foreground noise level. The construction of foreground model is particularly useful to enhance the segmentation performance in such aspect.

3.3 Swimmer detection with filtering scheme at nighttime

Robust operation of automated surveillance systems at night, particularly in aquatic environments, faced additional challenges posed by glare and reflections from the use of artificial lightings such as floodlights. This issue could be cast as a camouflaged target detection problem as swimmers are clearly hidden from view amid the glare of the pool's reflections. In the literature, the two-level thresholding scheme proposed in [3] to compensate missing pixels of camouflaged targets is generally confined to environments with static backgrounds. In [11], a method based mainly on using temporal information has been proposed; unfortunately crucial and useful spatial information is neglected in this method.

In this section, a filtering scheme is described which efficiently utilizes both spatial and temporal information to compensate the value of a partly camouflaged target by generating a 'pseudo color' based on a newly defined concept termed 'motion frequency'. The main idea is for each pixel to be firstly classified as one of three types as quantified by its motion frequency, based on which an appropriate filter operation is invoked accordingly.

3.3.1 Pixel classification based on motion frequency

Through extensive statistically analysis of our video data over long periods of time, we observed that: i) most background pixels has low pixel color fluctuations, ii) the main

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parts of a moving foreground, dynamic parts of the background and reflections caused by artificial lighting have a medium level of pixel color fluctuations, and iii) fast moving parts of the foreground and video acquisition noise both have high pixel color fluctuations. Such observation suggests that the rate of pixel color fluctuations provides a useful property to be exploited for the classification of pixels into various classes with different characteristics.

Let us first propose the quantifiable term motion frequency which measures the color fluctuation property of a pixel. Let $f_{x,y}^{i-N}$, $f_{x,y}^{i-N+1}$, ..., $f_{x,y}^{i}$ be intensity values of pixel at spatial location (x,y) for $N+1$ consecutive frames (although we are working with color video frames, we have found that pixel intensities provide sufficient indication of color fluctuations caused by real and apparent motion), with the sampling rate of M video frames per second. We define *motion frequency* of this pixel as:

$$\varpi_{x,y}^{i} = \frac{1}{M}\sum_{j=0}^{N} P_{x,y}^{j}, \quad P_{x,y}^{j} = \begin{cases} 1, & \text{if } D_{x,y}^{i,j} > Th, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

$D_{x,y}^{i,j}$ is the frame difference of the pixel between frames i and frame j, i.e., $D_{x,y}^{i,j} = \left|f_{x,y}^{i} - f_{x,y}^{j}\right|$; and Th is a robust threshold that determines the occurrence of a significant change in intensity. Clearly motion frequency is high if there are many such consecutive changes in pixel intensities and vice versa. Such motion frequency analysis also shares some of the effective properties of the linear prediction model given in [12]

3.3.2 Filtering scheme

This simple concept for quantifying intensity fluctuations is found effective in classifying pixels to one of the three classes mentioned above. Two robust thresholds, γ_a and γ_b which are fixed for all operational conditions need to be specified, after which pixels are routed to the appropriate filters according to the following scheme:

$$y_{x,y}^{i} = \begin{cases} s_{x,y}^{i}; & \varpi > \gamma_b, \\ g_{x,y}; & \gamma_a < \varpi \le \gamma_b, \\ f_{x,y}^{i}; & \varpi \le \gamma_a. \end{cases} \tag{13}$$

Setting γ_a and γ_b at one thirds and two thirds of N/M produce the best results as tested over long operational hours. For $\varpi \le \gamma_a$, no filtering action was implemented to avoid unnecessary blurring on the background. For $\varpi > \gamma_a$ and $\gamma_a < \varpi \le \gamma_b$, $s_{x,y}^{i}$ and $g_{x,y}$ are the respective outputs of a mean and the proposed 'pseudo' color compensation filters with the respective definition as follows.

i) Mean filtering, $s_{x,y}^{i}$: We apply a mean filter operating on pixels within a $(2n+1)\times(2n+1)$ window to yield output:

$$s_{x,y}^{i} = \frac{1}{(2n+1)^2}\sum_{j=-n}^{n}\sum_{k=-n}^{n} f_{x+j,y+k}^{i}. \tag{14}$$

Typically, choose $n = 1$, i.e., a 3×3 window, with consistent performance achieved to remove undesirable noise.

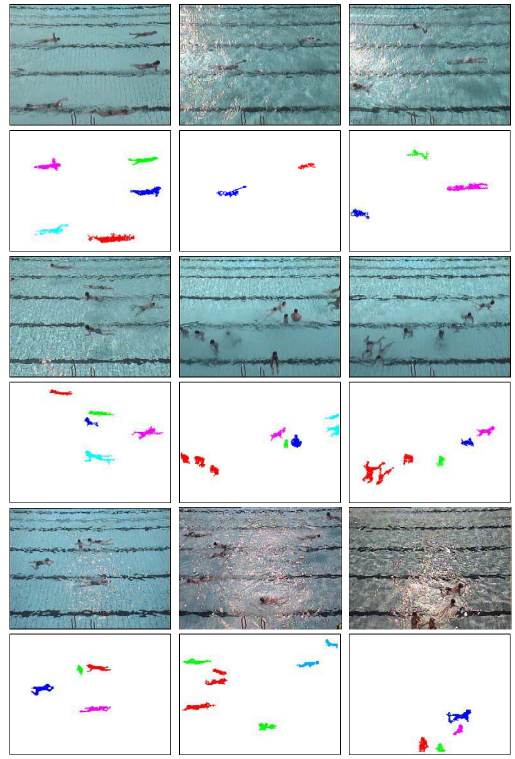


Figure 5: The segmentation of swimmers on sample frames captured at different time intervals from 9am to 8pm. Odd rows: Samples of scene captured; Even rows: Segmented swimmers using our proposed algorithm.

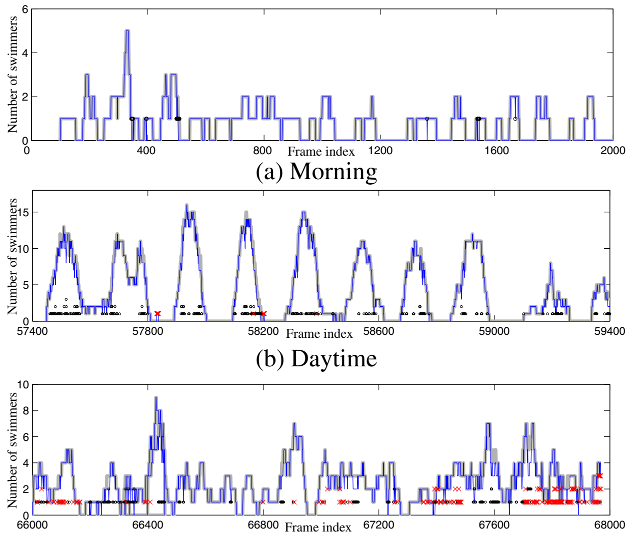
ii) A proposed 'pseudo' color compensation filter, $g_{x,y}$: It aims to generate a 'pseudo color' for pixel located in the 'compensation region' by performing a spatio-temporal mean filtering. Let W be a set of mean filtered pixels for $N+1$ successive frames and $\forall(x,y) \in C$, where C is the set of pixels belonging to compensated regions, i.e., $W_{x,y} = \{s_{x,y}^{i+p}\ ;\ -N \le p \le N\}$. Therefore, the output of the color compensation filter $g_{x,y}$ is defined by:

$$g_{x,y} = R_{x,y}. \tag{15}$$

where $R_{x,y}$ denotes the average color of elements in $W_{x,y}$ except the white color of water reflection.

4 Experimental results

A real-time surveillance system has been set up on trial for continual monitoring at an outdoor olympic-sized public swimming pool for the past one year. The system consists of a network of overhead cameras mounted surrounding the pool such that the entire pool is covered by the combined view of the cameras. Running real-time at 4 frames/second throughout day and night, the system operates under numerous outdoor situations amid glare, rain and nighttime reflections.



(a) Morning

(b) Daytime

(c) Nighttime

Figure 6: Results on three selected video streams with length of 2000 frames for each at different time intervals within the same day. (Legends: Lighter line shows the ground truth of the number of swimmers in a frame; Darker line shows the number of detected swimmers; "∘" indicates false negative; "×" indicates false positive.)

4.1 Operations over an entire day (and night)

Figures 5 and 6 present consecutive detection of swimmers at the outdoor pool from 9am to 8pm on one typical weekday. Samples of image (arranged from left to right, top to bottom) depict numerous typical scenes and challenges encountered to provide a sampling of the difficult task involved. Various challenging conditions, such as sunlight reflections in the morning, high swimmer densities during swimming lessons and the presence of distinct shadows, all combined to make swimmer detection a very difficult task especially amid the dynamic conditions with partially hidden swimmers. Despite the challenges, the proposed algorithms show promising results, demonstrating its robustness to operate under various real situations over very long operational hours. The second image of the first column shows a successful example of the system being able to immediately detect a swimmer moving in from the left. In the third image, the system manage to detect a swimmer moving within a region full of lighting reflections. Even for crowded situations during swimming lessons, the system achieved a good detection rate with some child swimmers being very small indeed.

The proposed algorithm was also evaluated using an objective measure to quantify its detection and false positive performances. Figure 6 depicts the obtained results on three typical video sequences of length 2000 frames each. Figure 6 (a) presents a typical scenario at the swimming pool during weekdays, recording a total of 1500 swimmers in the video sequence of 2000 frames. 1473 swimmers have been correctly detected, achieving a detection rate of 98.20% and showing that the detected swimmer curve follows the ground truth closely. The 1.80% of miss detection cases are mainly due to the low contrast between these swimmers and the background and the extremely small size of certain swimmers which makes detection very difficult. Meanwhile, the proposed algorithm is also robust in adapting to the dynamic movements of the background, achieving a 0% false positive rate in this experiment.

Figure 6 (b) presents results obtained during a swimming lesson. This is a more complicated scenario compared to that in (a), involving more swimmers with some of them being small children and a more hostile background due to greater water disturbances from increased swimmer activities. Naturally, there would be an increase in the false negative and false positive rates. In this experiment, containing 8974 swimmers over 2000 frames, the proposed algorithm maintains a satisfactory result by recording a 4.2130% false negative rate and a 0.1658% false positive rate. Occasionally, parts of the lane divider have been mis-classified as foreground due to the system being misled by violent water disturbances.

The algorithm also shows consistent detection rate when operating at nighttime despite strong reflections caused by the use of floodlights at night. It achieved a 3.9186% false negative and a 5.7473% false positive while detecting 5359 swimmers over 2000 frames. A higher false positive compared with that of the daytime sequence has been mainly due to the misclassification of certain reflections regions as swimmers.

4.2 Handling specular reflection at nighttime

Figure 7 provides a good comparison of the detection performance of the system without and with the color compensation scheme based on the proposed motion frequency at nighttime frames. As could be seen in the 2^{nd} row of Figure 7, i.e., without the preprocessing scheme, it was difficult to correctly extract parts of swimmers that were occluded by water reflections. Additionally, some apparent movements at the lane dividers due to water ripples have been mistakenly classified as foreground regions. The detection results presented in the 3^{rd} row show much improvement with the incorporation of the proposed color-compensation filtering scheme, demonstrating greater foreground noise immunity while achieving satisfactory segmentation results.

4.3 Performance comparison

We also benchmarked the proposed algorithm against the well known W4 system [1] and the mixture of Gaussian (MoG) method by Stauffer et. al. [2] Results for the W4 and MoG methods are obtained after carefully fine tuning various parameters used in these methods. The higher efficiency of our algorithm in detecting small and weak vis-

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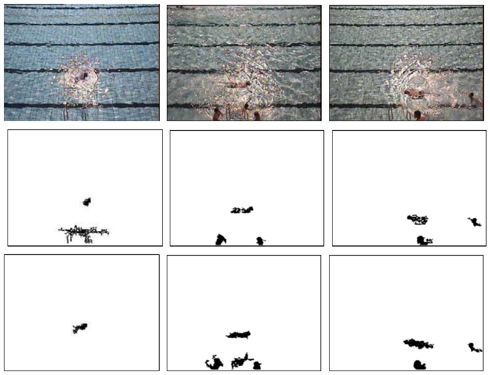

Figure 7: A comparison of detected swimmers without and with filtering. 1^{st} row: Sample frames of a typical nighttime sequence of an outdoor pool; 2^{nd} row: Detected swimmers without filtering; 3^{rd} row: Detected swimmers with the proposed spatio-temporal filtering scheme.

ible swimmers as well as maintaining a low false positive error (i.e., mis-classification of background elements to be foreground) has yielded better performance compared with W4 and MoG methods as shown in Figure 8. The figure demonstrate successful examples of the system being able to detect small swimmers at various positions on the frame while maintaining a low false positive error; meanwhile, both W4 and MoG have failed to achieve. It is understandable that this may not be a fair comparison. We would like to highlight here that the current state-of-the-art algorithms are generally found inadequate and the framework may not be straightforwardly extended to such new problem.

Table 1 presents the evaluation in terms of objective measure to quantify the detection and false positive performances for two different scenarios: a typical less crowded scenario for the first sequence and a crowded with high activity scenario during a swimming lesson for the second sequence. It illustrates satisfactory results for our algorithm with only slight deterioration in performance when tested on the second sequence compared with substantial deterioration observed for W4 and MoG methods.

5 Conclusion

This paper presented various unique insights into effective foreground detection amid highly dynamic environments and demonstrated the operation of the developed algorithms on a live public swimming pool environment over long periods of time. A unique region-based background model and a corresponding foreground detection scheme that captures long-range spatial dependencies within dynamic backgrounds have been developed. Superior foreground detection was achieved without increasing false positive rates through an implementation of a thresholding-with-hysteresis scheme. Finally, a pre-processing filtering

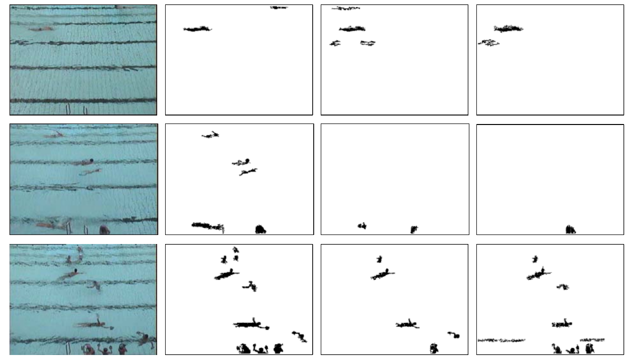

Figure 8: A comparison of swimmer segmentation on a typical scene captured. 1^{st} column: Samples of scene captured; 2^{nd} to 4^{th} columns: Results obtained using our proposed algorithm, W4 and MoG, respectively.

approach based on newly defined concepts of motion frequency and color compensation has been proposed to recover pixels of foreground hidden within specular reflections. Promising results demonstrated the robustness of the developed methodology when appiled on a very challenging live system operating under hostile conditions.

	Seq. #1			Seq. #2		
	TP(%)	TN(%)	FP(%)	TP(%)	TN(%)	FP(%)
Our algorithm	99.74	0.26	0.78	97.26	2.74	0
W4 [1]	89.87	10.13	8.57	62.22	37.78	0.12
MoG [2]	96.88	3.12	1.82	73.44	26.56	5.99

Table 1: Comparison of detection and false positive performances. (Legends: TP represents True Positive; TN represents True Negative; FP represents False Positive.)

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